

Practical Assignment - 2

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Roll No : 05

Subject : 304 Open Source Web Development

Course : MSc.ICT

Semester : 3

Question 1

Practical Assignment: Blog Management REST API with Authentication (Node.js, Express, MongoDB, EJS)

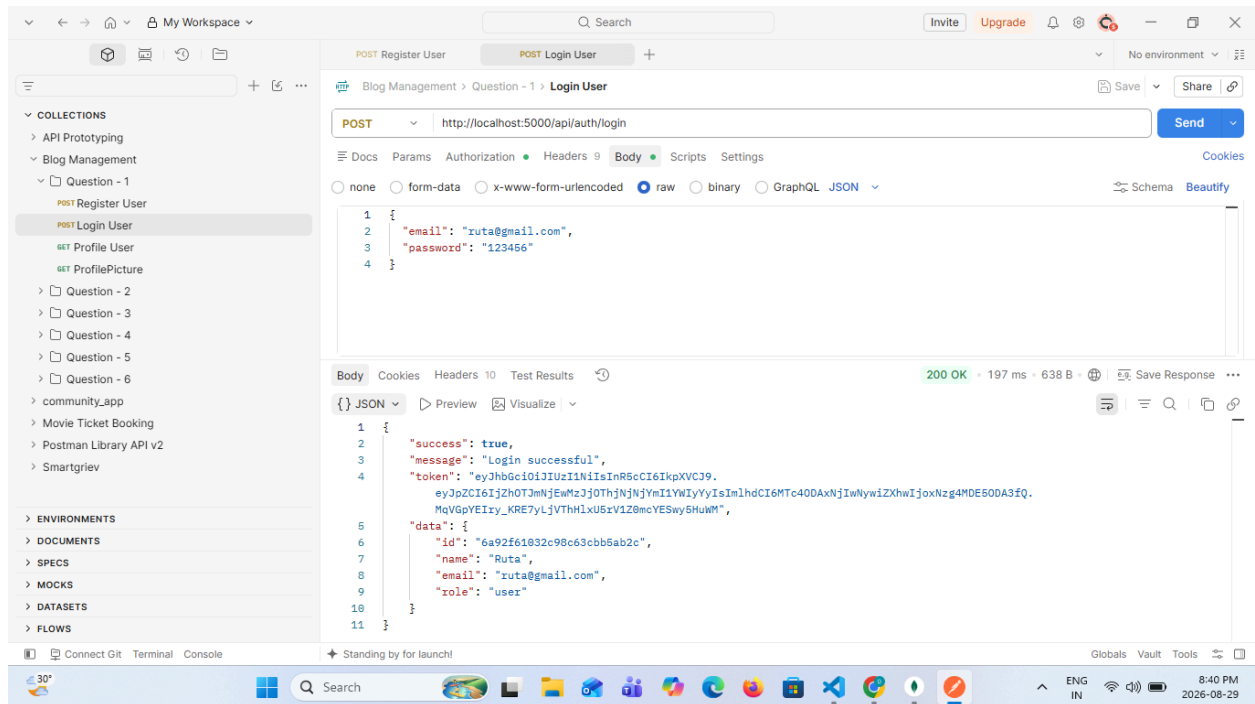
Scenario: A startup wants to build the backend and a basic server-rendered interface for a personal blogging platform where registered users can sign up, log in, and manage their own blog posts. Only the author of a post (or an admin) should be able to edit or delete it. You have been asked to build this system.

Build an application using Node.js, Express.js, MongoDB, Mongoose, and EJS (for rendering views) that implements the following:

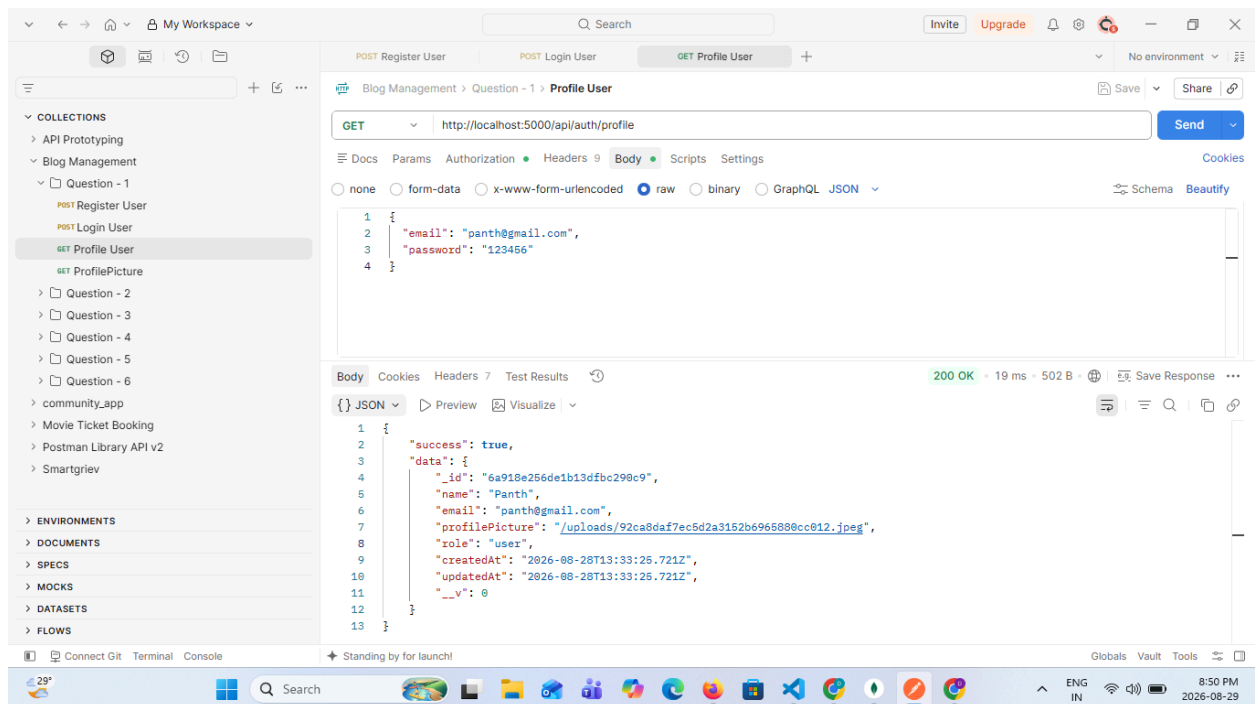
1. User Authentication

- POST /api/auth/register – Register a new user (name, email, password) with the password hashed using bcrypt.

- POST /api/auth/login – Authenticate a user and return a JWT token.



- GET /api/auth/profile – A protected route that returns the logged-in user's profile.



2. Blog Post CRUD API (protected routes, linked to the logged-in author)

- POST /api/posts – Create a new blog post (title, content, tags, published status).

The screenshot shows the Postman interface with a POST request to `http://localhost:5000/api/posts`. The request body is a JSON object:

```
{  "title": "My Second Blog Post",  "content": "This is my second blog post created using Node.js, Express.js and MongoDB.",  "tags": ["Node.js", "Express", "MongoDB"],  "published": true}
```

The response is a 201 Created status with a JSON body:

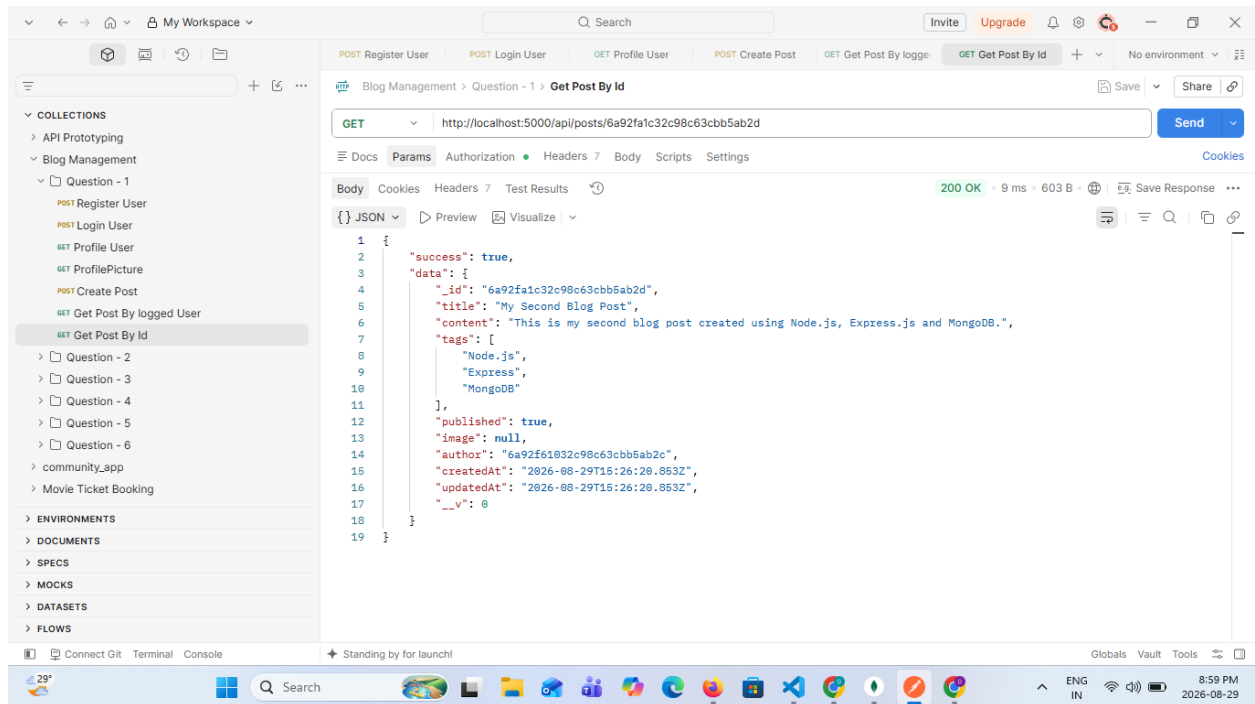
```
{  "success": true,  "message": "Post created successfully",  "data": {    "title": "My Second Blog Post",    "content": "This is my second blog post created using Node.js, Express.js and MongoDB.",    "tags": [      "Node.js",      "Express",      "MongoDB"    ],    "published": true,    "image": null,    "author": "6a92f61032c98c63cbb5ab2c",    "_id": "6a92fa1c32c98c63cbb5ab2d",    "createdAt": "2026-08-29T15:26:20.853Z",    "updatedAt": "2026-08-29T15:26:20.853Z"  }}
```

- GET /api/posts – Retrieve all blog posts belonging to the logged-in user.

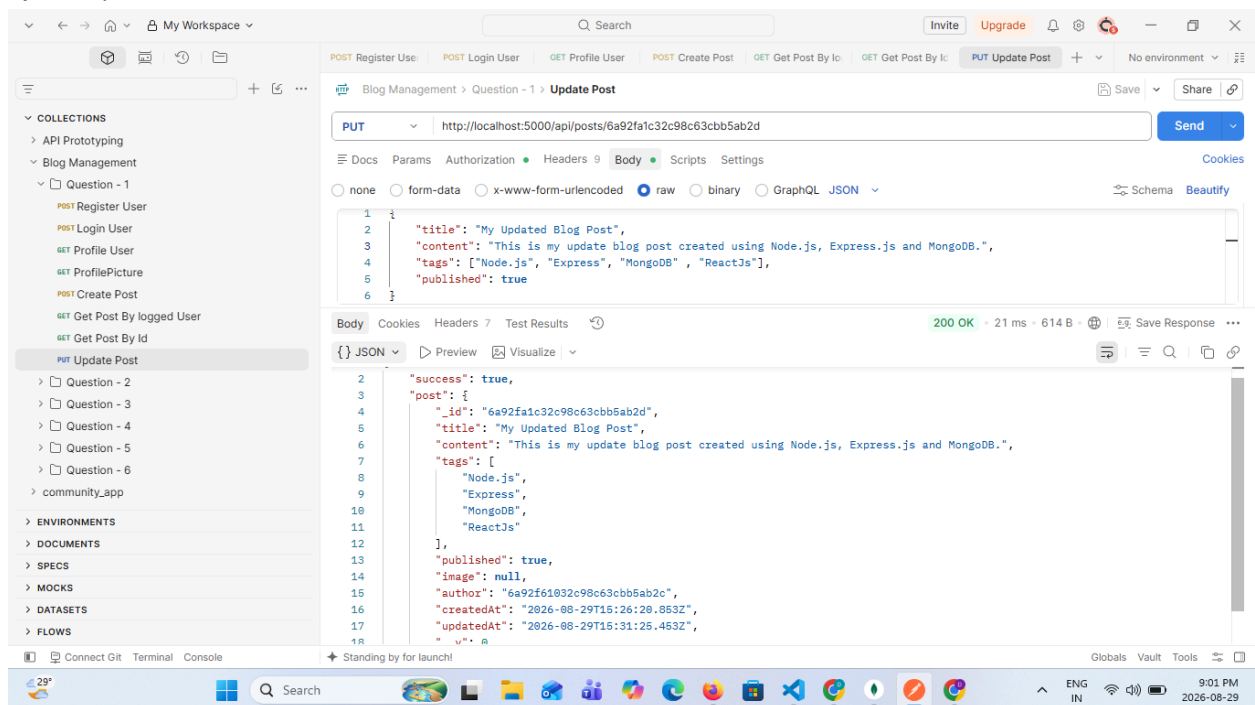
The screenshot shows the Postman interface with a GET request to `http://localhost:5000/api/posts`. The response is a 200 OK status with a JSON body:

```
{  "success": true,  "count": 1,  "data": [    {      "_id": "6a92fa1c32c98c63cbb5ab2d",      "title": "My Second Blog Post",      "content": "This is my second blog post created using Node.js, Express.js and MongoDB.",      "tags": [        "Node.js",        "Express",        "MongoDB"      ],      "published": true,      "image": null,      "author": "6a92f61032c98c63cbb5ab2c",      "createdAt": "2026-08-29T15:26:20.853Z",      "updatedAt": "2026-08-29T15:26:20.853Z",      "__v": 0    }  ]}
```

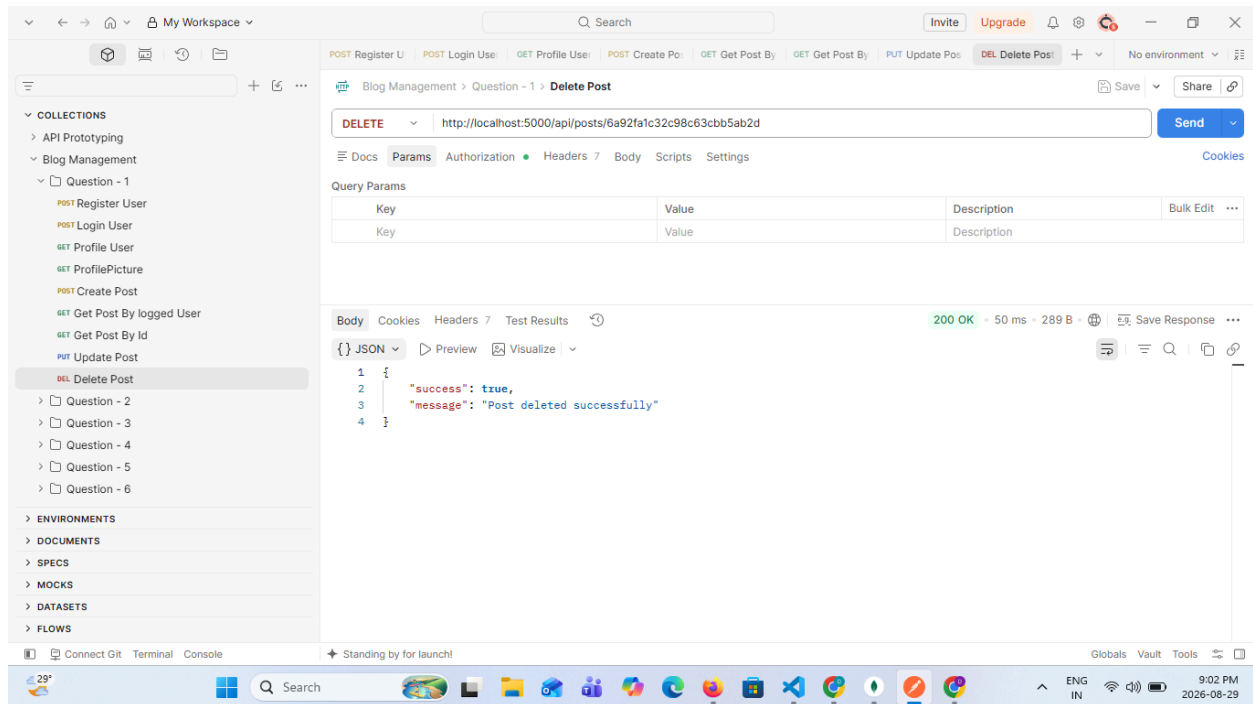
- GET /api/posts/:id – Retrieve a single blog post by ID.



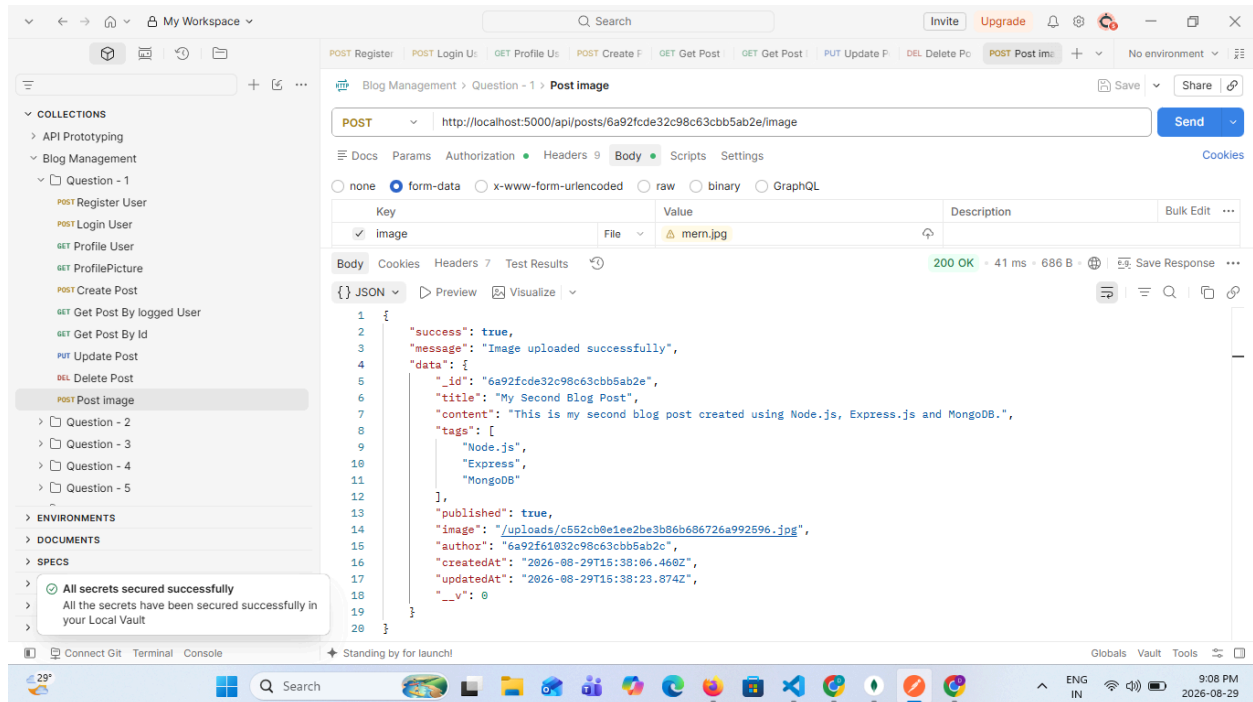
- PUT /api/posts/:id – Update a blog post by ID (only the author or an admin can update).



- DELETE /api/posts/:id – Delete a blog post by ID (only the author or an admin can delete).

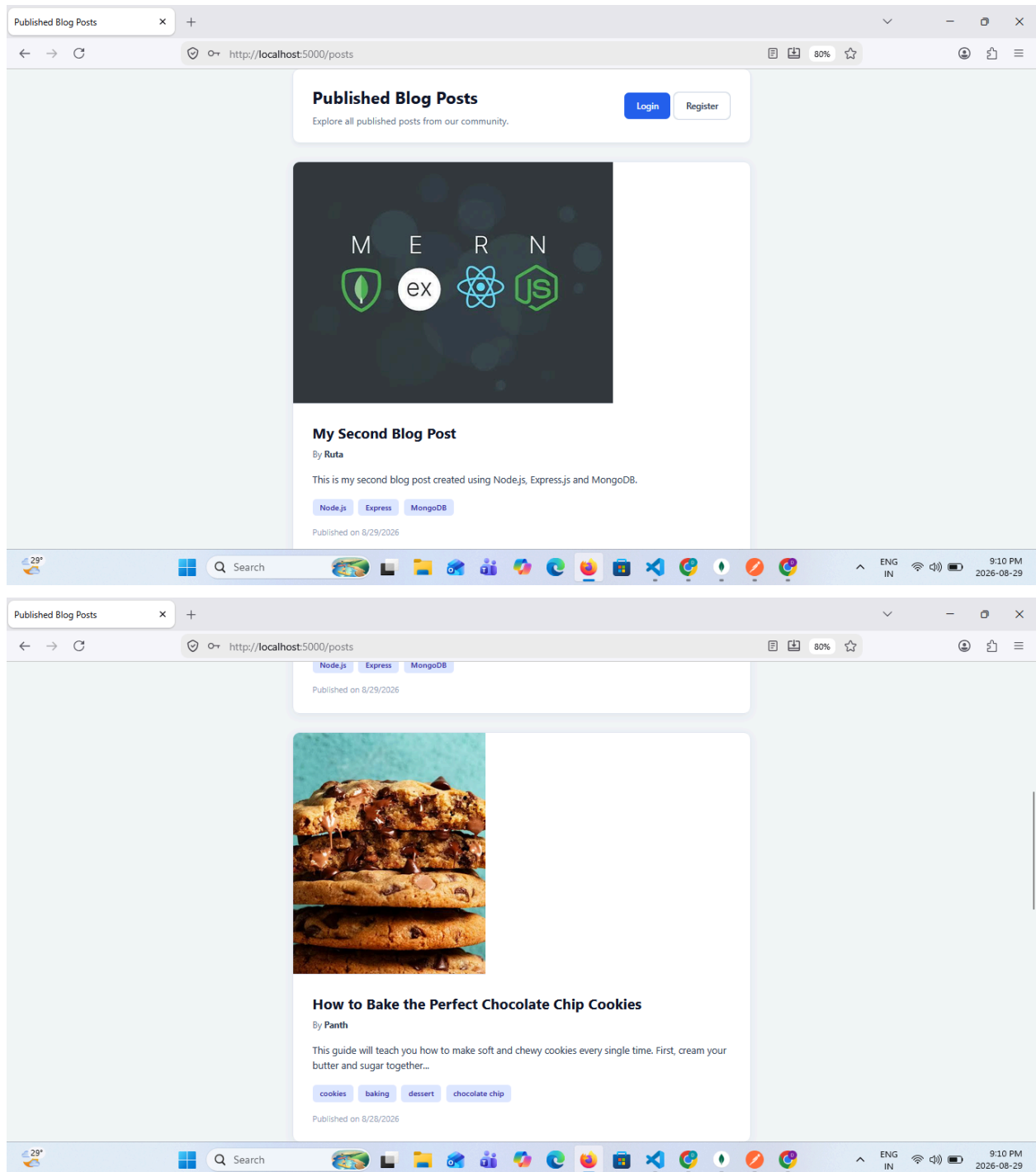


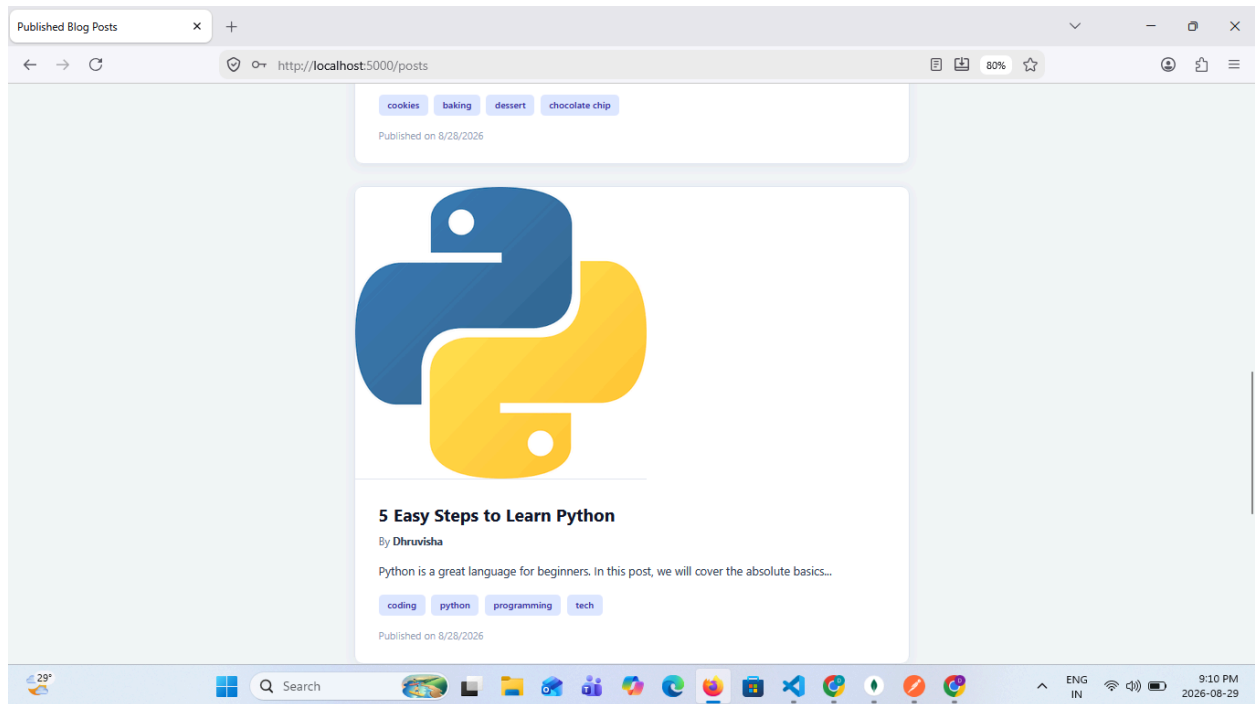
- POST /api/posts/:id/image – Upload or replace a featured image for a blog post (multipart/form-data) using Multer.



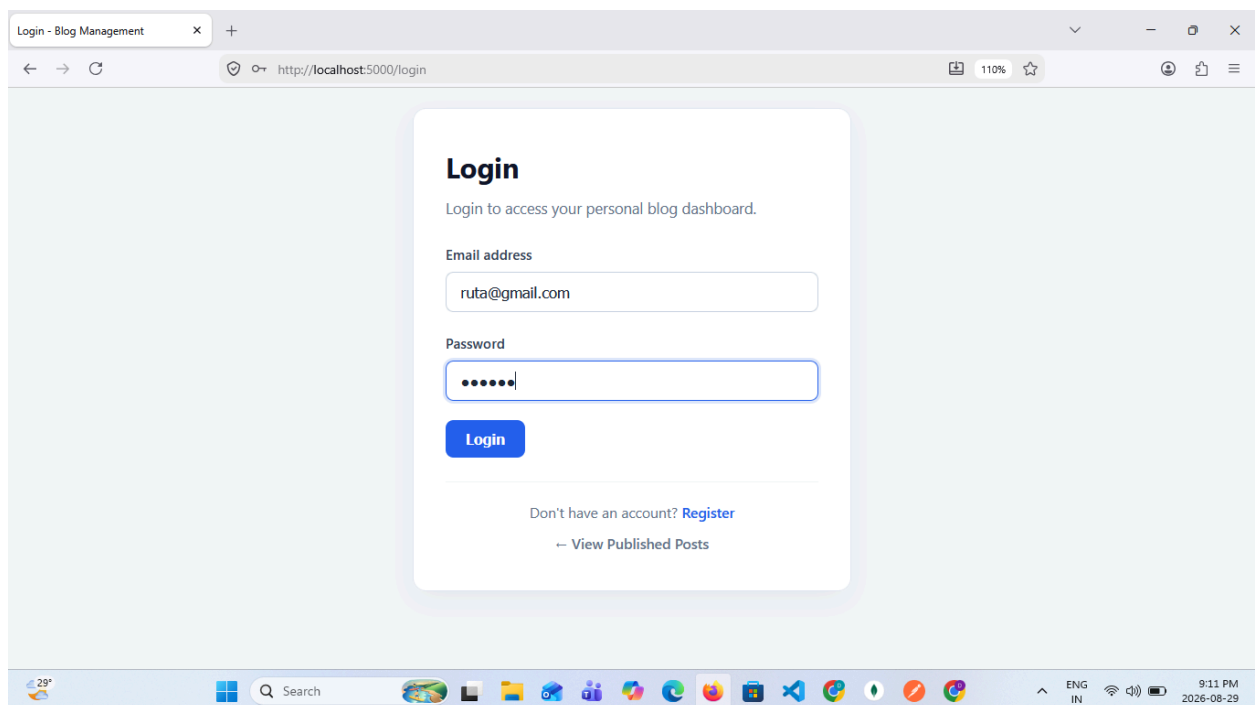
3. EJS Views

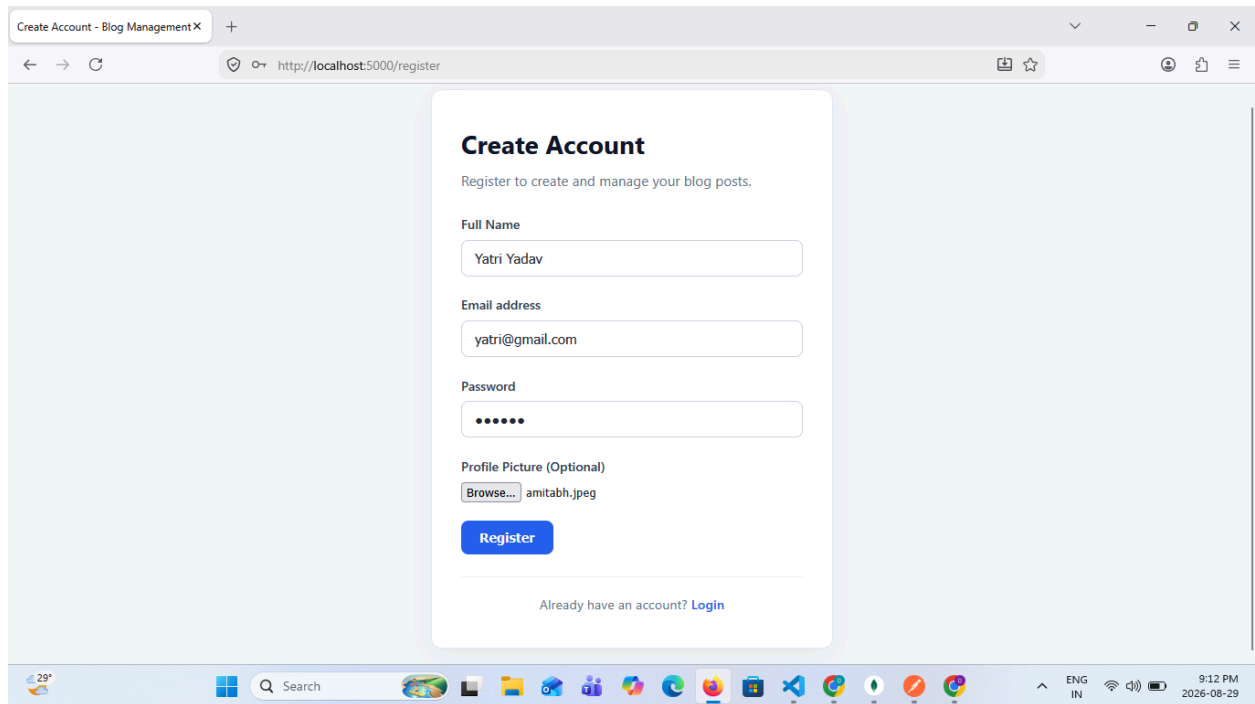
- Render an EJS page (views/posts.ejs) that displays all published blog posts in a styled list.



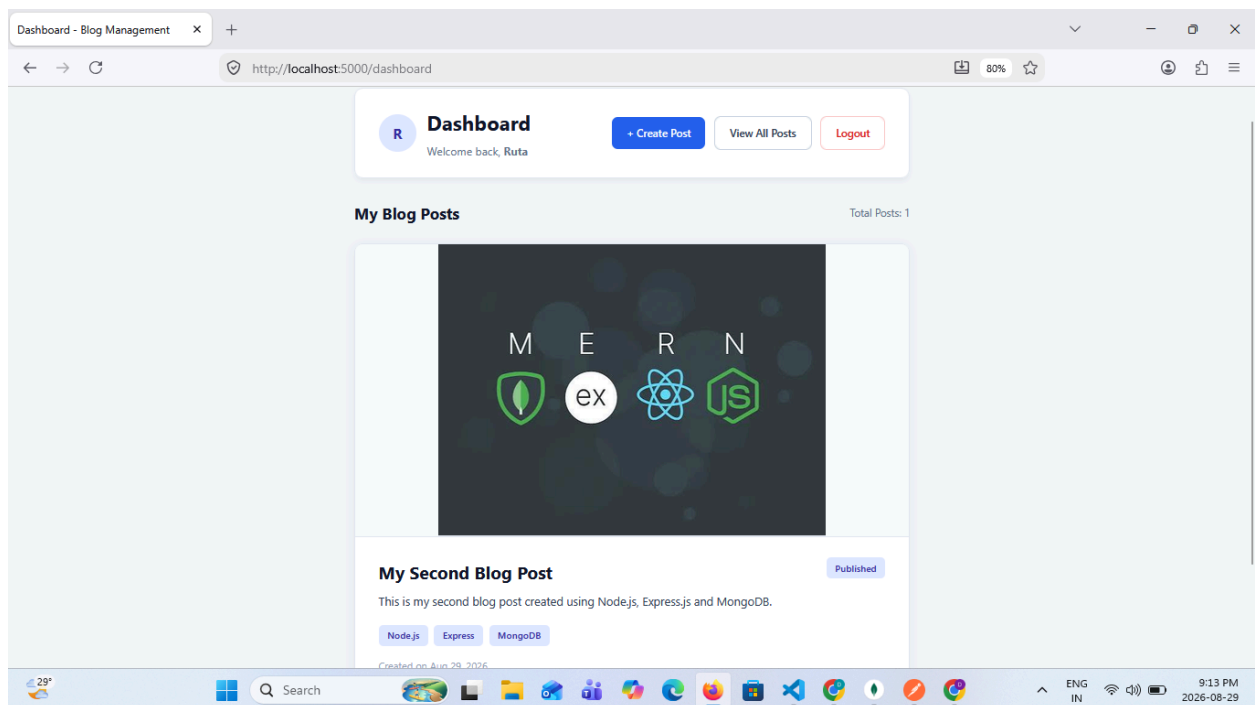


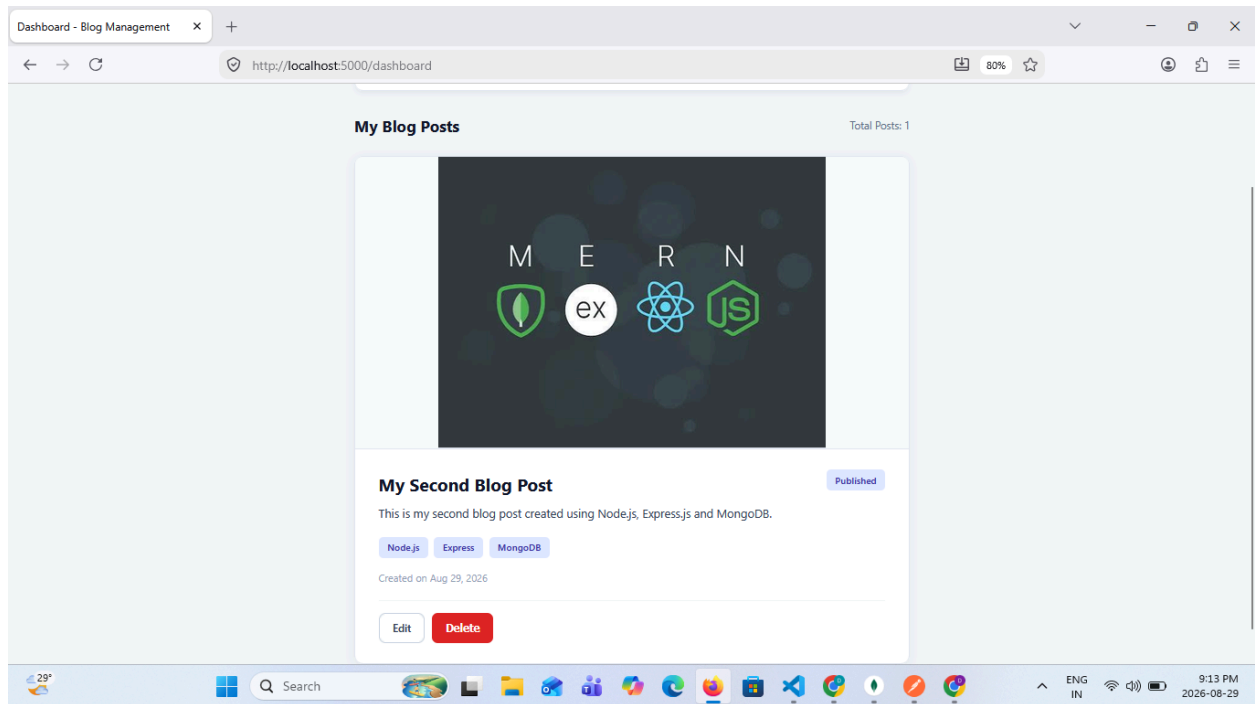
- Render an EJS page (views/login.ejs) with a login form that submits to your login route.





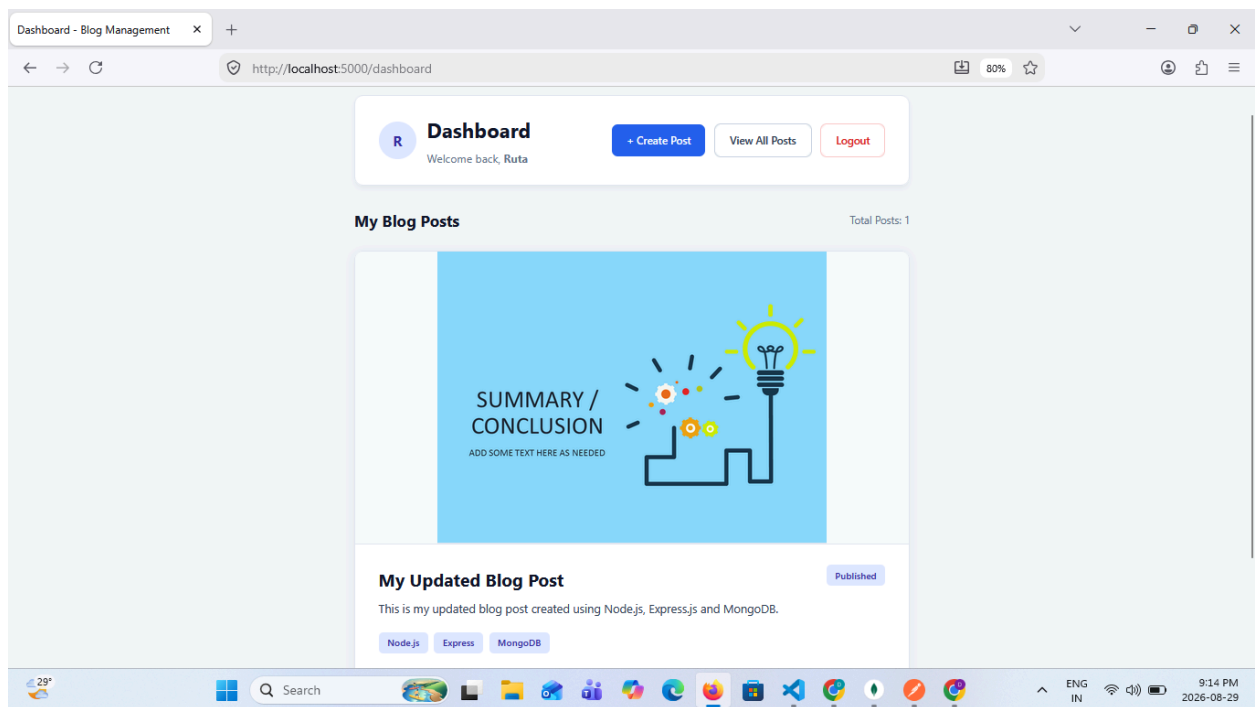
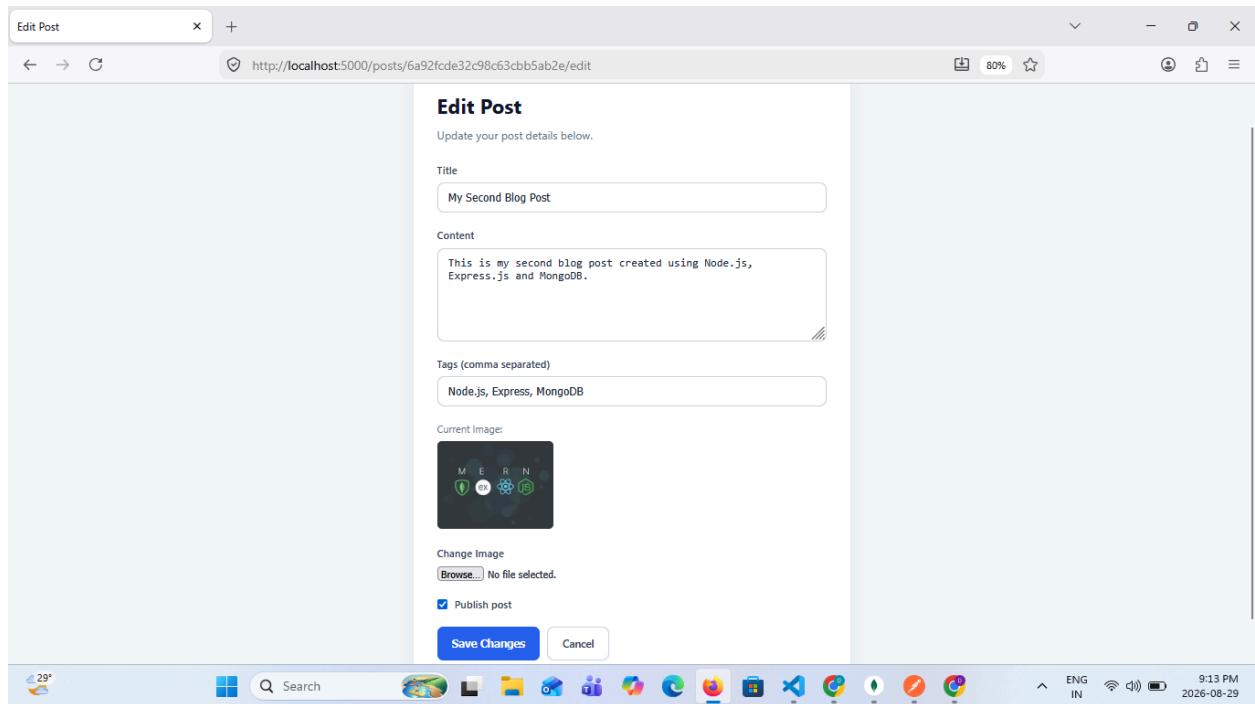
- Render an EJS page (views/dashboard.ejs, protected) showing the logged-in user's own posts with edit/delete links, visible only after login.





4. File Uploading

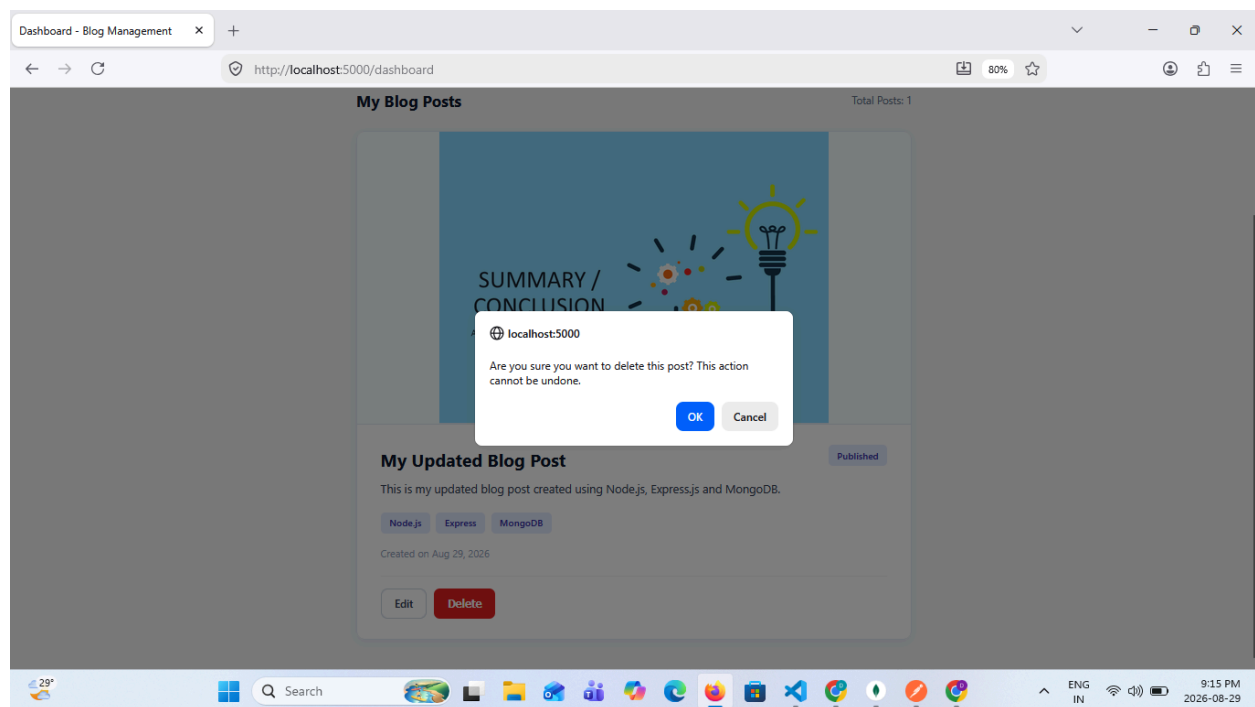
- Use Multer to handle multipart/form-data uploads for a blog post's featured image and/or a user's profile picture.
- Restrict uploads to image MIME types (jpeg, png, webp) and enforce a maximum file size (e.g., 2MB); reject and return a clear error for anything else.
- Store uploaded files in a local /uploads directory (or a cloud service such as Cloudinary/AWS S3) and save only the resulting file path/URL on the Post or User document.
- Serve uploaded images as static assets and render them in views/posts.ejs and views/dashboard.ejs.
- Delete the associated image file from storage when a post (or its image) is deleted or replaced.



5. Security

- Use Helmet to set secure HTTP headers, and configure CORS to only allow trusted origins.

- Apply rate limiting (e.g., express-rate-limit) on /api/auth/login and /api/auth/register to mitigate brute-force and credential-stuffing attacks.
- Validate and sanitize all request input (e.g., with express-validator or Joi) to guard against NoSQL injection and malformed payloads.
- Keep secrets (JWT signing key, MongoDB URI, upload credentials) in environment variables via dotenv and never commit them to source control (.env.example only).
- Set a reasonable JWT expiry, sign tokens with a strong secret, and reject/refresh appropriately on expiry.
- Validate uploaded file types on the server (not just by extension) and never trust the client-supplied filename when saving to disk.

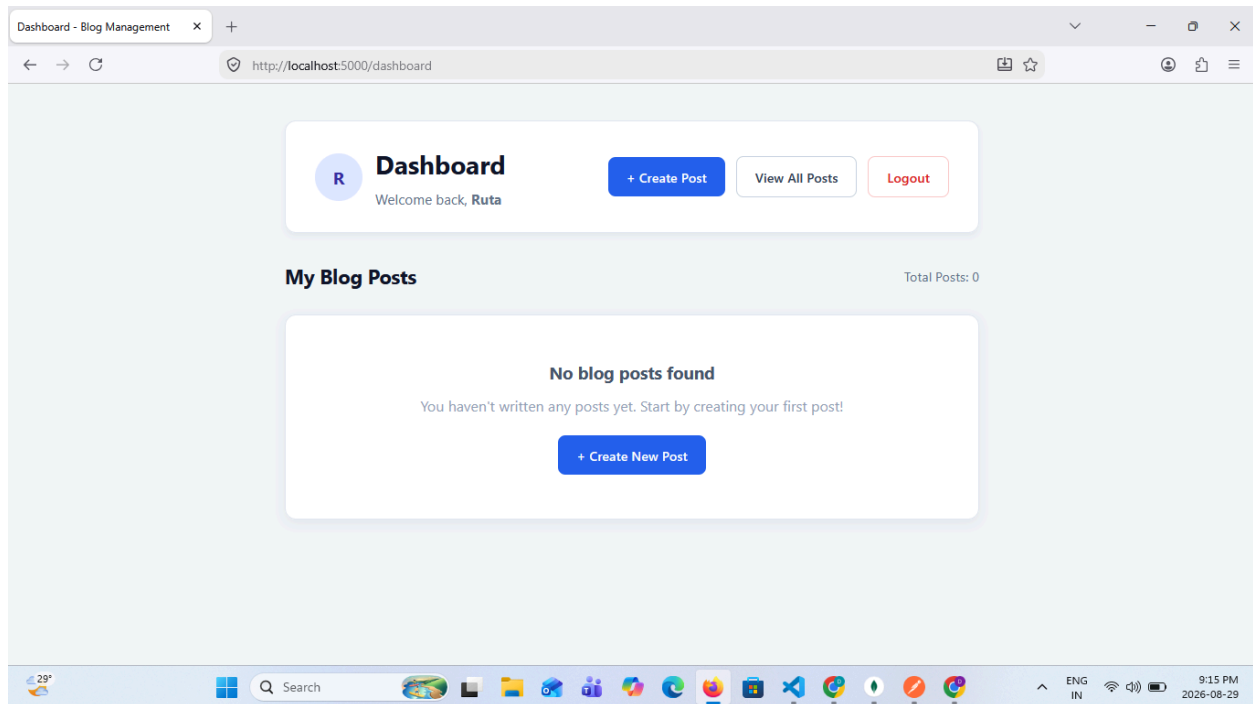


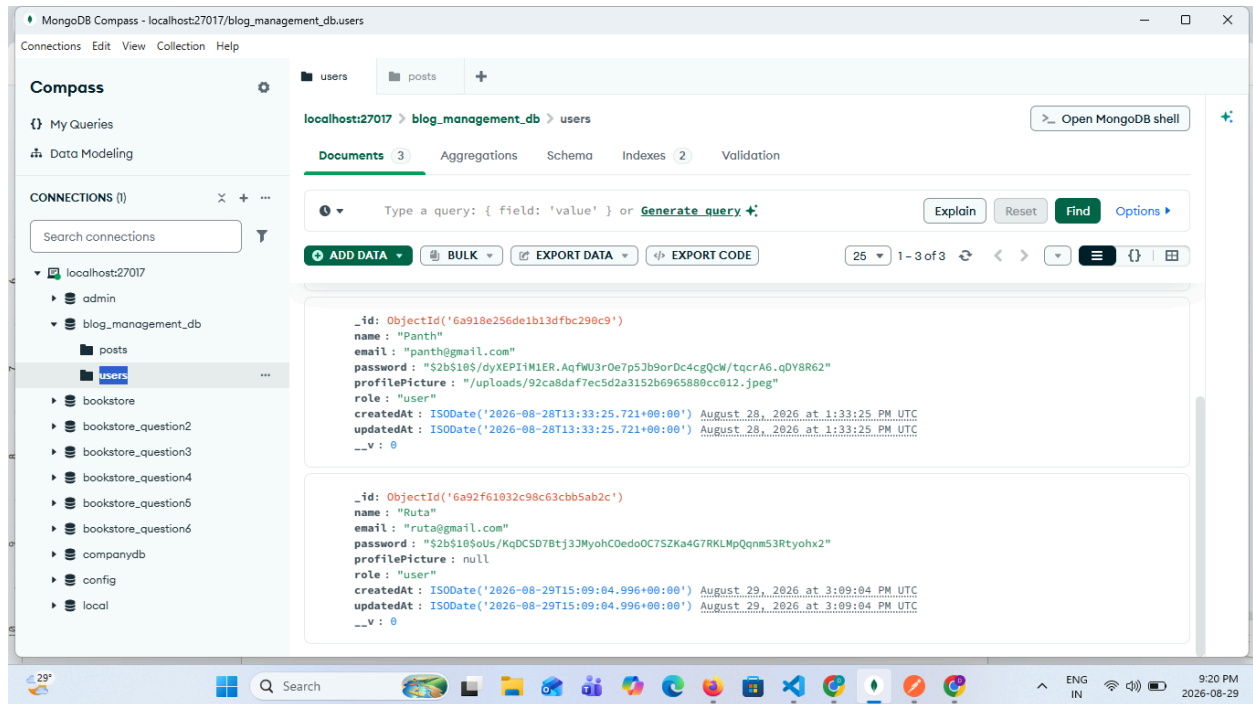
6. Requirements

- Use bcrypt to hash and compare passwords — never store plain text passwords.
- Use JWT middleware to protect all post-related API and dashboard routes.
- Use Mongoose schemas to model User and Post, with each post referencing its author via ObjectId and each user having a role field (user/admin).
- Ensure a user can only view/edit/delete their own posts, unless they are an admin.
- Return appropriate HTTP status codes and consistent JSON responses for the API routes.
- Never trust client input for authorization decisions — always re-check ownership/role server-side, even if the UI hides edit/delete links.

- Apply the same Security requirements above (Helmet, rate limiting, input validation, env-based secrets) across both the API and file-upload routes.

Deliverable: A working Node.js/Express project with proper folder structure (models, routes, controllers, middleware, views, uploads), a README.md explaining setup and usage (including a .env.example listing required environment variables), and example API requests (curl or Postman) demonstrating registration, login, all blog post CRUD operations, and image upload, plus screenshots or a demo of the EJS pages.



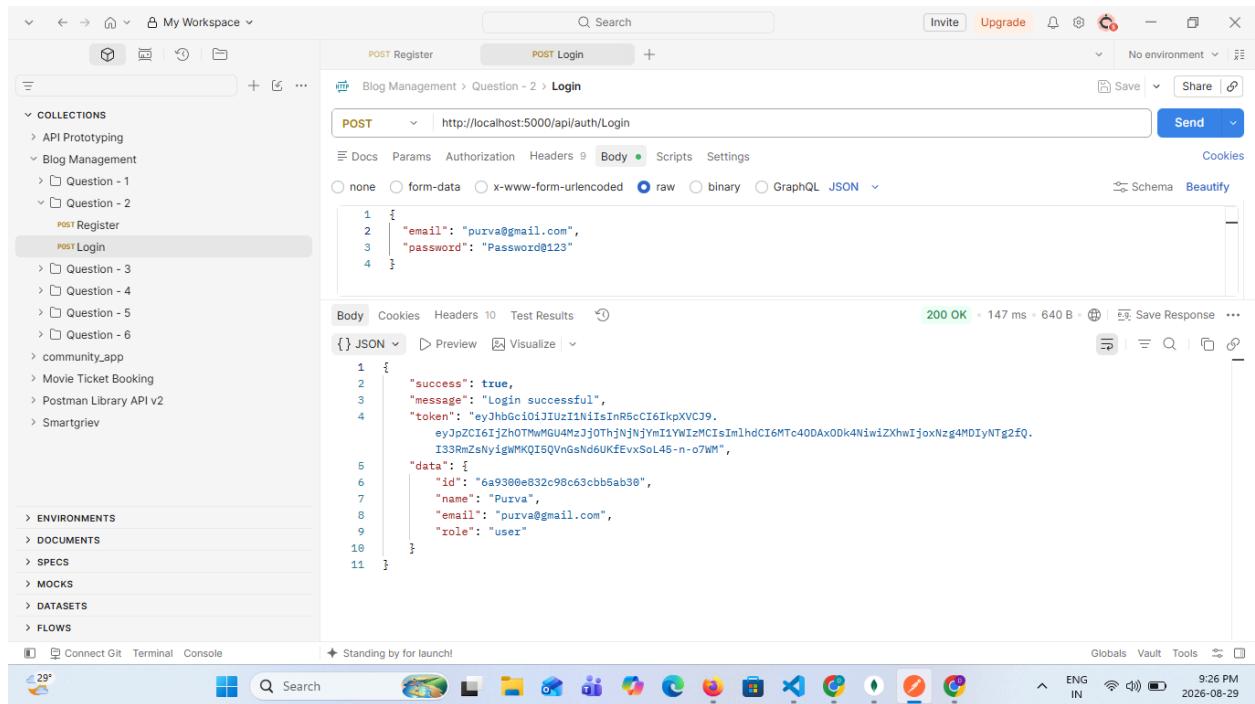


Question 2

Secure Signup for Bookstore Customers

Scenario: A new customer is creating an account on the online bookstore to start placing orders for books.

Task: Write a program that hashes the customer's password using bcrypt before storing it in MongoDB, and write the corresponding logic to verify the password when the same customer logs in later.

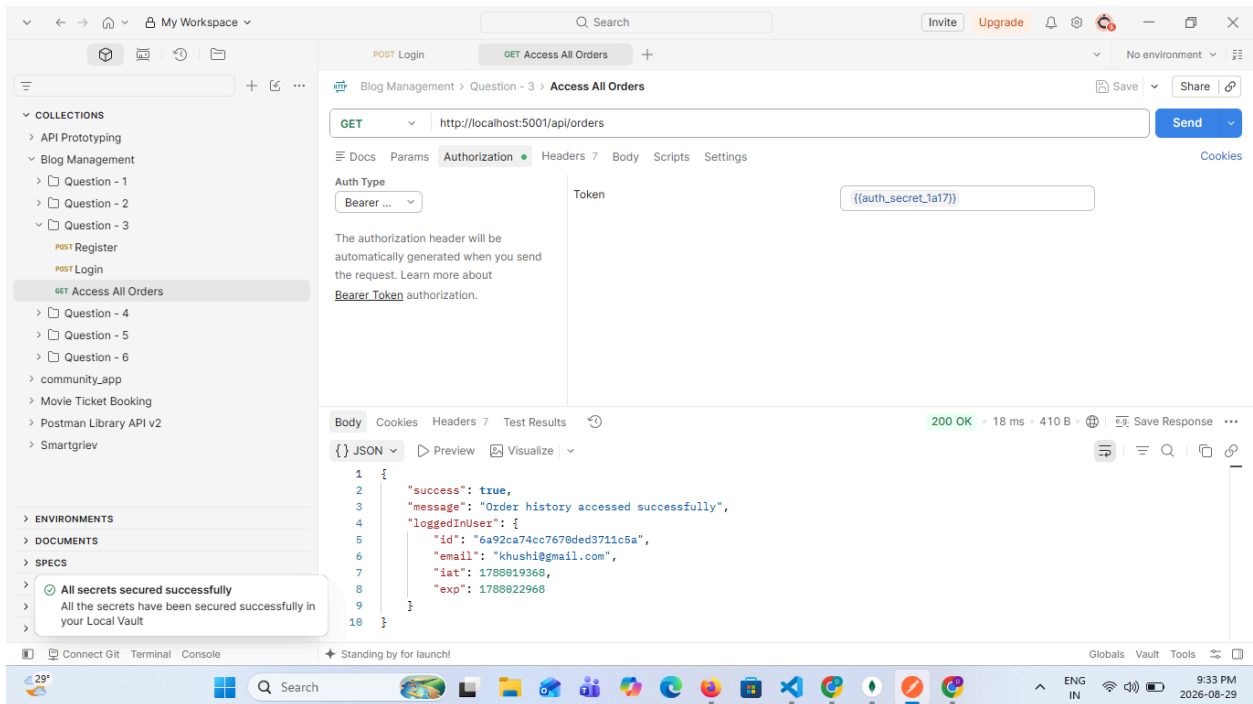
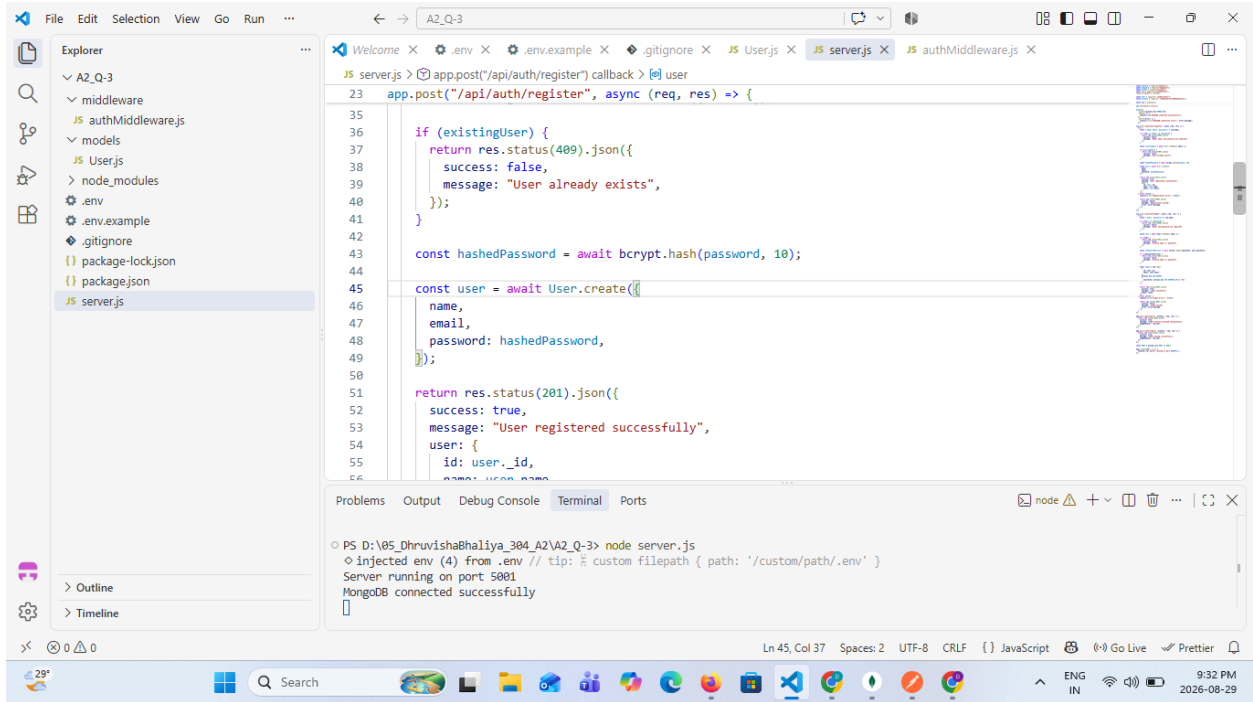


Question 3

Issuing a Digital Membership Token on Login

Scenario: Once a customer logs in successfully, the store must issue them a token (their digital membership pass) so they don't need to log in again for every request, such as viewing their orders or checking out.

Task: Write a program to generate a JWT token on successful login, and a middleware to verify this token before allowing access to any protected route (e.g., placing an order or viewing order history).

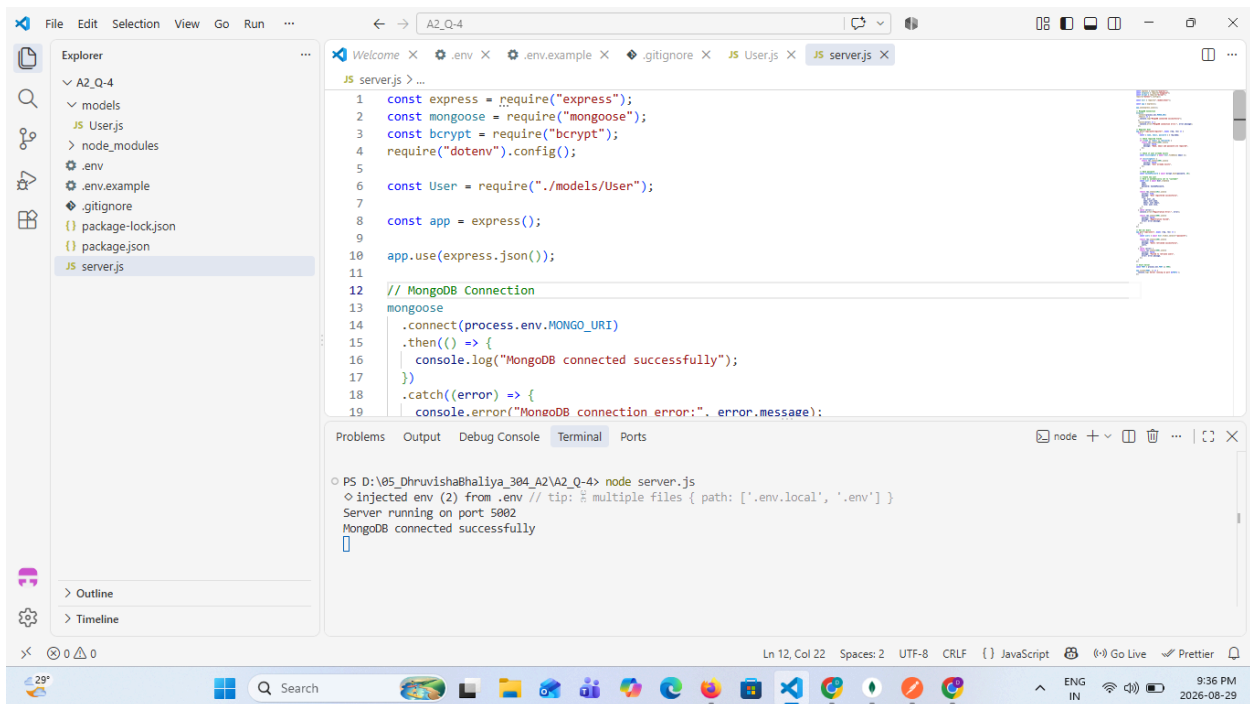


Question 4

Defining Customer and Admin Roles

Scenario: The bookstore platform has two kinds of accounts — regular Customers who browse and buy books, and Admins who manage inventory, pricing, and all customer orders.

Task: Write a Mongoose User schema with a role field ('customer' / 'admin') that defaults to 'customer', so every new signup is treated as a regular customer unless explicitly promoted to admin.



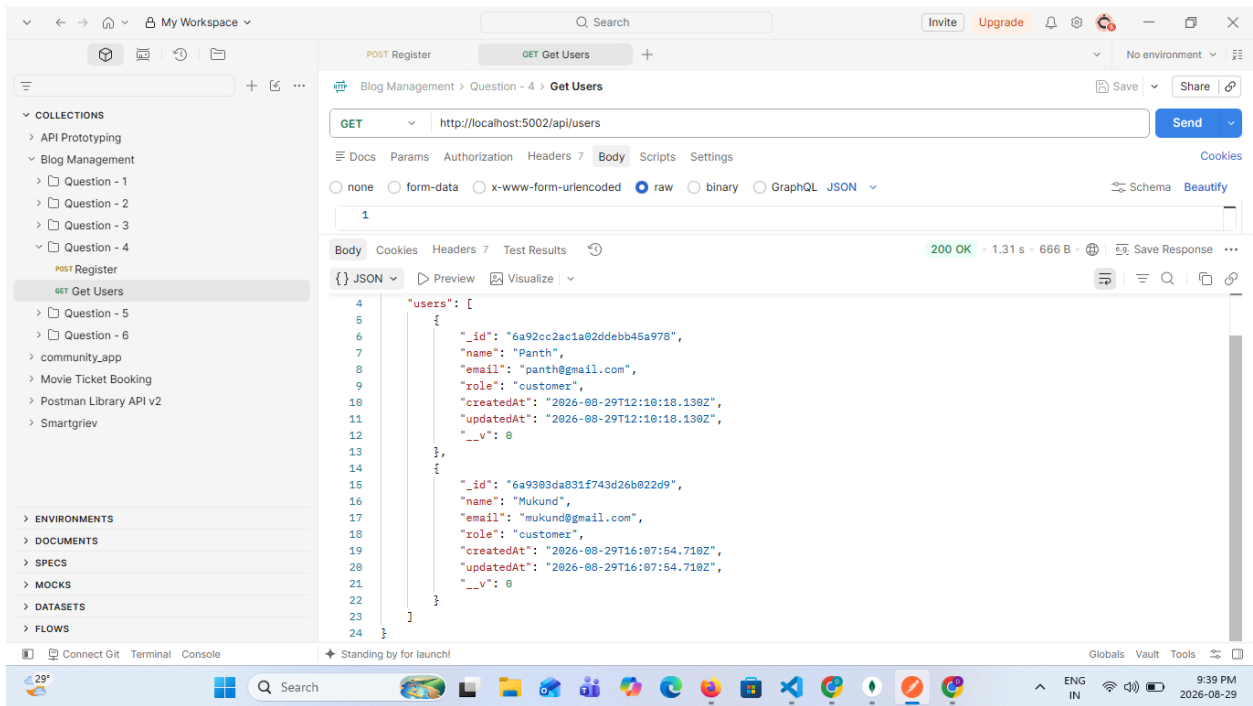
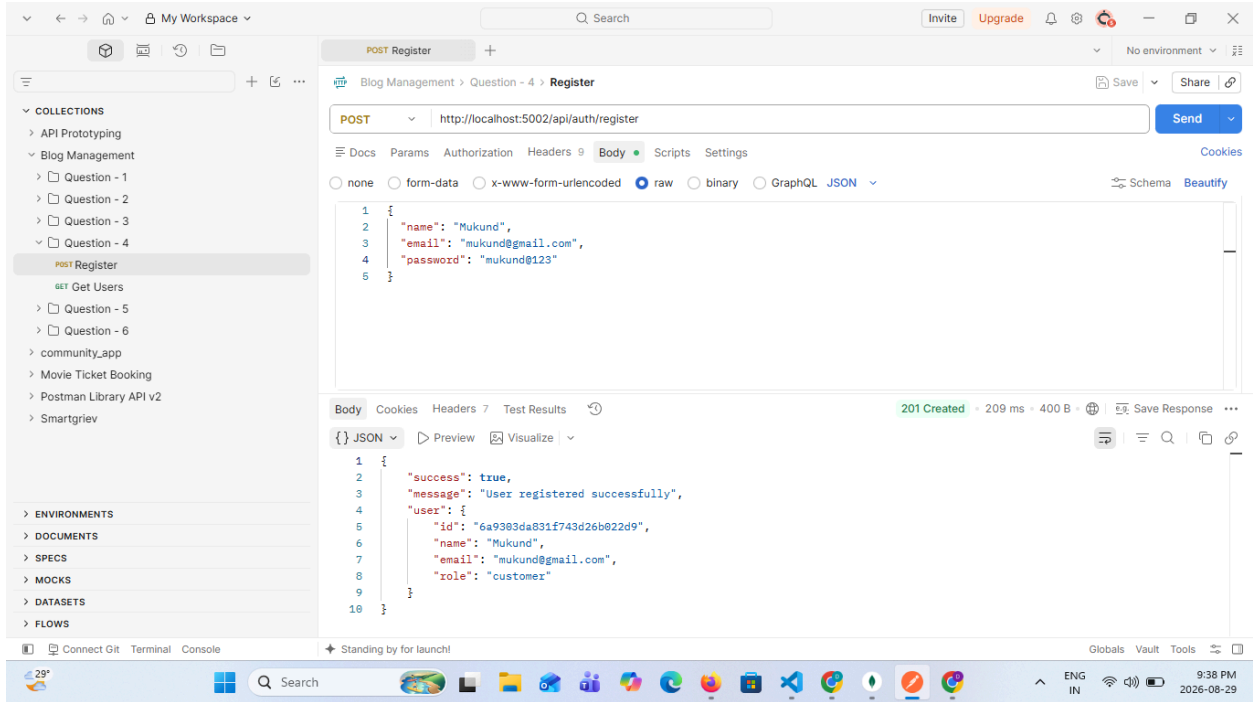
The screenshot shows a Visual Studio Code editor window with the following components:

- Explorer:** A file tree on the left showing the project structure. The file `server.js` is selected.
- Editor:** The main workspace displays the content of `server.js`. The code includes imports for `express`, `mongoose`, and `bcrypt`, along with a Mongoose connection and a basic Express server setup.
- Terminal:** The bottom panel shows the output of running `node server.js`. It indicates that the server is running on port 5002 and that MongoDB connected successfully.

```
1 const express = require("express");
2 const mongoose = require("mongoose");
3 const bcrypt = require("bcrypt");
4 require("dotenv").config();
5
6 const User = require("./models/User");
7
8 const app = express();
9
10 app.use(express.json());
11
12 // MongoDB Connection
13 mongoose
14   .connect(process.env.MONGO_URI)
15   .then(() => {
16     console.log("MongoDB connected successfully");
17   })
18   .catch((error) => {
19     console.error("MongoDB connection error:", error.message);
20   });
21
22 app.listen(5002, () => {
23   console.log("Server is running on port 5002");
24 });
```

Terminal Output:

```
PS D:\05_DhruvishaBhaliya_304_A2\A2_Q-4> node server.js
[dotenv] injected env (2) from .env // tip: multiple files { path: ['.env.local', '.env'] }
Server running on port 5002
MongoDB connected successfully
```

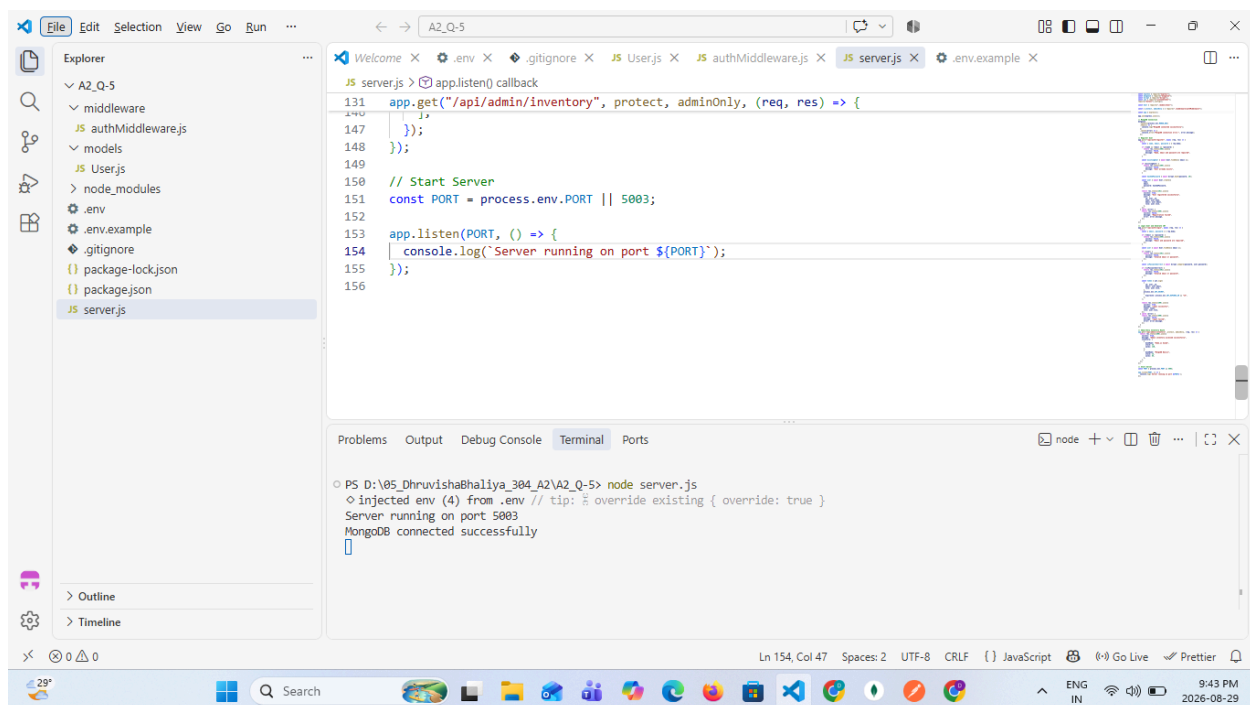


Question 5

Admin-Only Access to the Inventory Dashboard

Scenario: The bookstore has an inventory management route (GET /api/admin/inventory) that lists stock levels and sales data for every book, meant only for store admins.

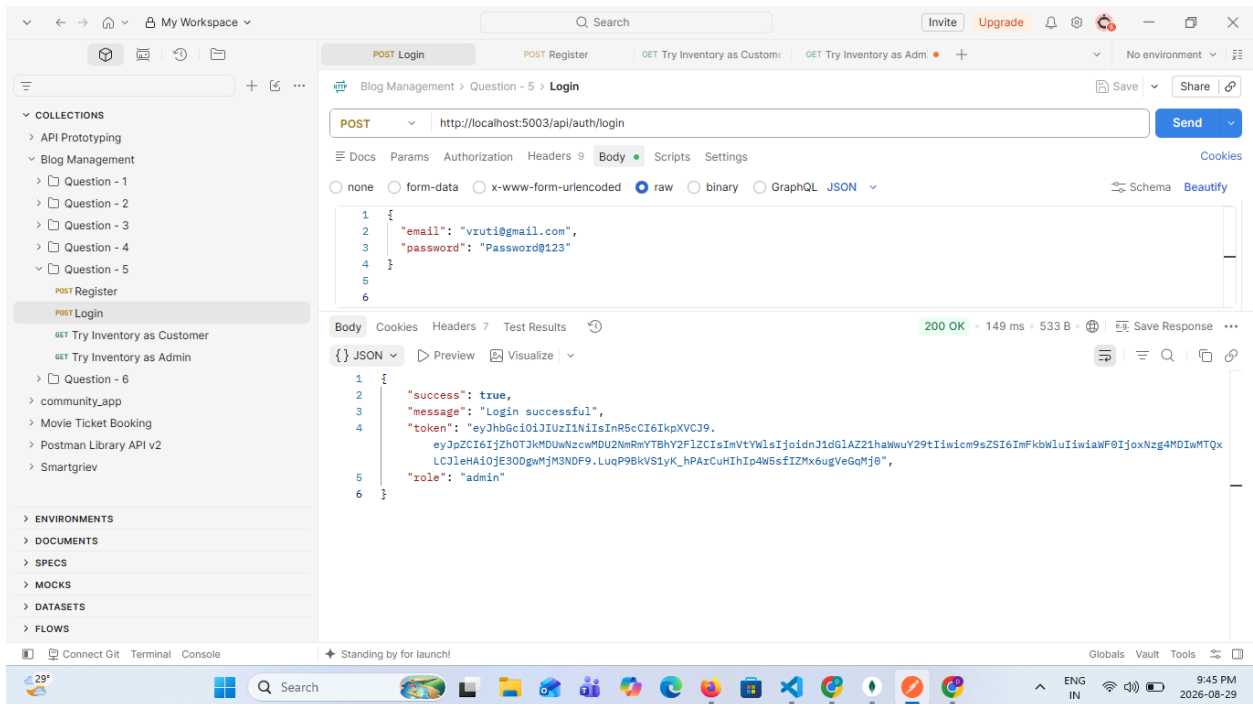
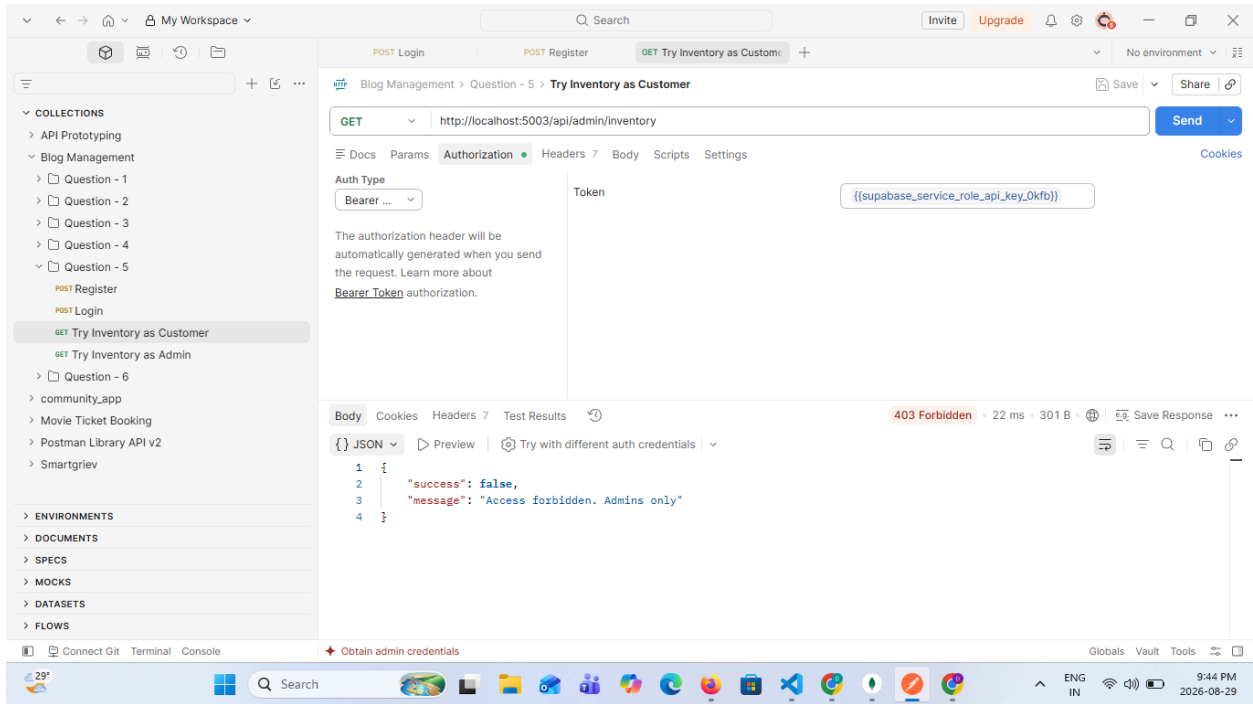
Task: Write an authorization middleware that checks the logged-in user's role and returns 403 Forbidden if the user is not an 'admin', preventing regular customers from accessing this route.

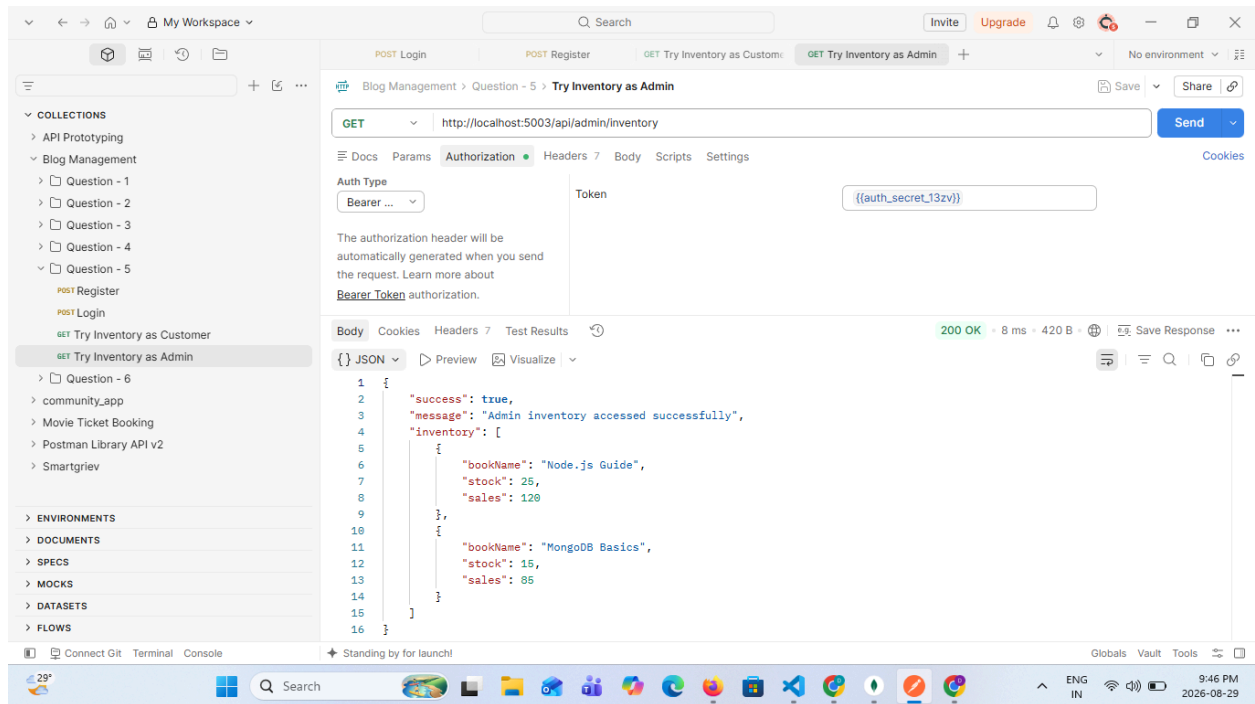


```
File Edit Selection View Go Run ... A2_Q-5
Welcome X .env X .gitignore X JS Userjs X JS authMiddleware.js X JS server.js X .env.example X
131 app.listen() callback
140
141 app.get("/api/admin/inventory", protect, adminOnly, (req, res) => {
142   //
143 });
144
145
146 // Start Server
147 const PORT = process.env.PORT || 5003;
148
149 app.listen(PORT, () => {
150   console.log(`Server running on port ${PORT}`);
151 });
152
153
154
155
156
```

Problems Output Debug Console Terminal Ports

```
PS D:\05_DhruvishaBhaliya_304_A2\A2_Q-5> node server.js
◊ injected env (4) from .env // tip: override existing { override: true }
Server running on port 5003
MongoDB connected successfully
```

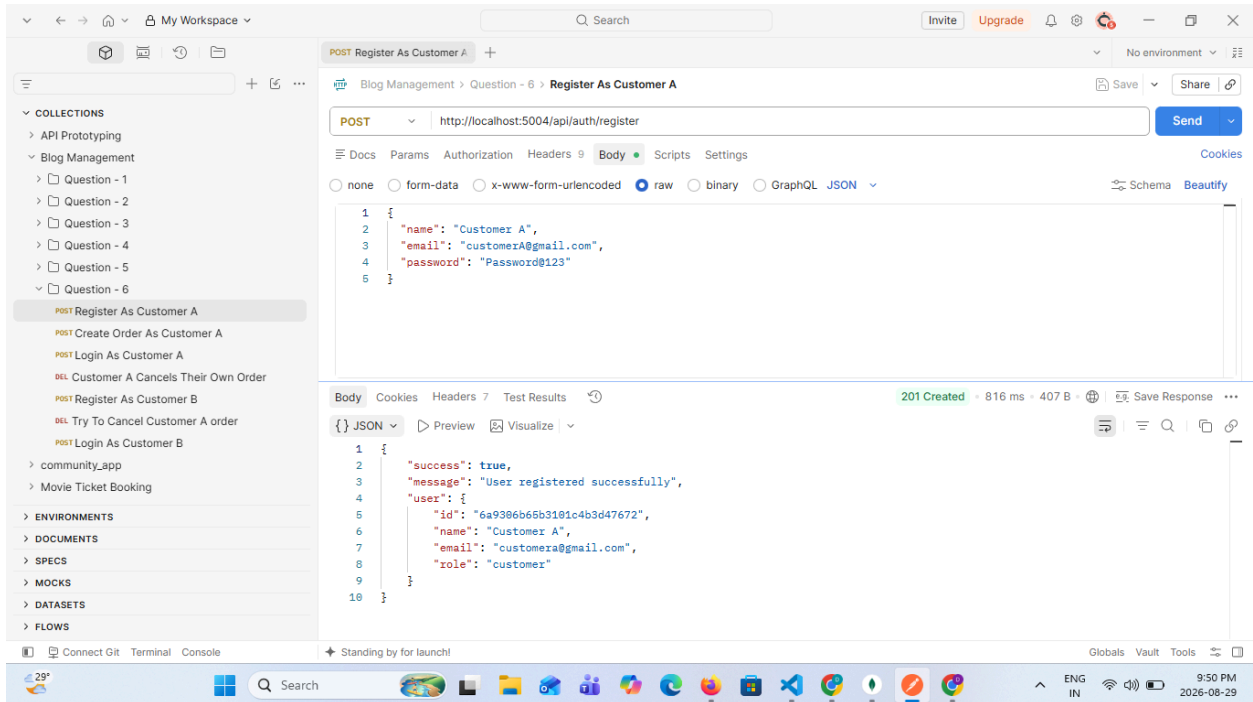
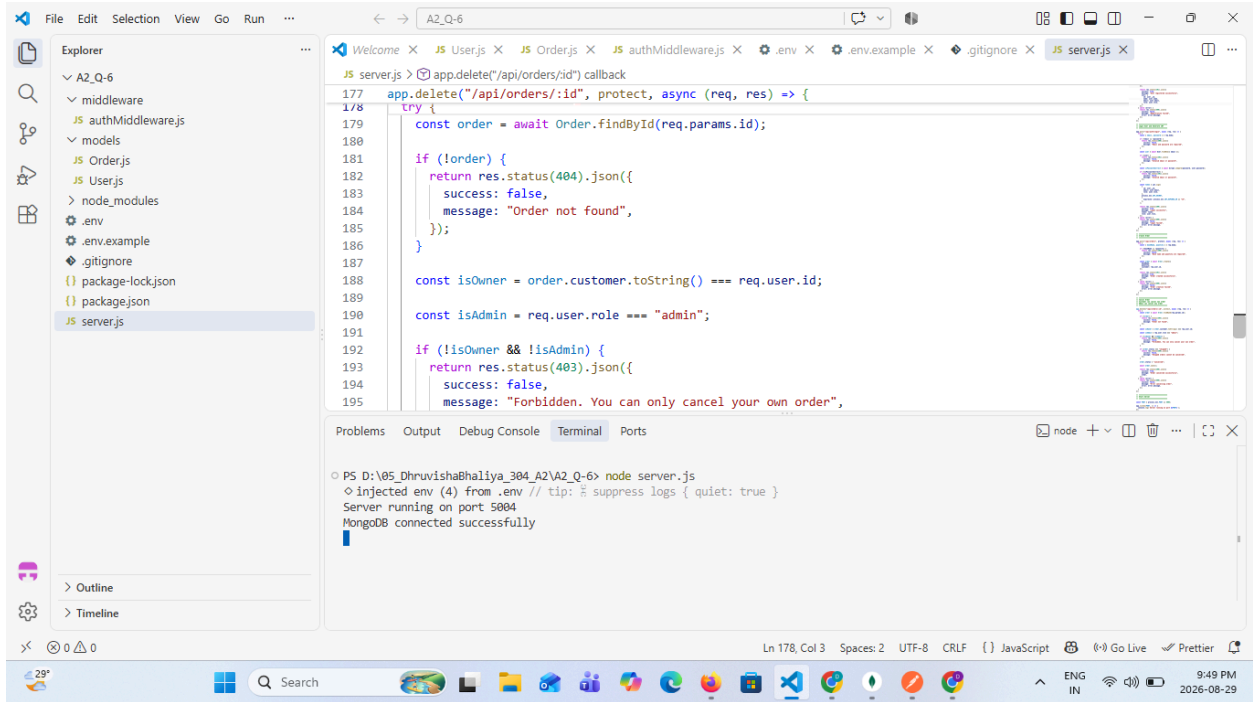


Question 6

Cancelling an Order — Customer or Admin Only

Scenario: A customer should be able to cancel their own order before it ships. Separately, an admin should be able to cancel any customer's order — for example, when a book goes out of stock.

Task: Write a DELETE `/api/orders/:id` handler that allows an order to be cancelled only if the logged-in user is the customer who placed the order, or holds the admin role, and returns 403 Forbidden otherwise.



My Workspace

Blog Management > Question - 6 > Customer A Cancels Their Own Order

DELETE http://localhost:5004/api/orders/6a9306e85b3101c4b3d47673

Auth Type: Bearer ... Token: {{supabase_service_role_api_key1311}}

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body: 200 OK - 23 ms - 522 B

```
{
  "success": true,
  "message": "Order cancelled successfully",
  "order": {
    "id": "6a9306e85b3101c4b3d47673",
    "bookName": "Node.js Guide",
    "quantity": 2,
    "customer": "6a9306e85b3101c4b3d47672",
    "status": "cancelled",
    "createdAt": "2026-08-29T16:20:56.692Z",
    "updatedAt": "2026-08-29T16:21:59.979Z",
    "__v": 0
  }
}
```

My Workspace

Blog Management > Question - 6 > Register As Customer B

POST http://localhost:5004/api/auth/register

Body: 201 Created - 244 ms - 407 B

```
{
  "success": true,
  "message": "User registered successfully",
  "user": {
    "id": "6a93073d5b3101c4b3d47674",
    "name": "Customer B",
    "email": "customerb@gmail.com",
    "role": "customer"
  }
}
```

